

The background of the slide features a close-up, shallow depth-of-field shot of several gold coins. Some coins are in sharp focus, showing numbers like '80', '60', '40', '25', and '30'. A dark, semi-transparent rectangular box is centered over the image, containing the course title and subtitle. A small red rectangular element is positioned at the top center of this dark box.

CPSC 4830

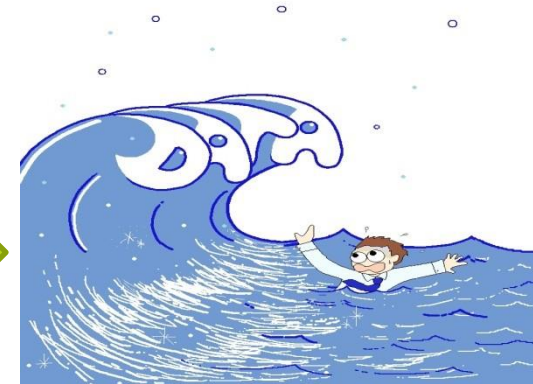
Data Mining for Data Analytics

Why Data Mining?

- ⌘ Explosive Growth of Data: from terabytes to petabytes

- ⌘ Data Collections and Data Availability

 - ⌘ Crawlers, database systems, Web, etc.



- ⌘ Sources

 - ⌘ Business: Web, e-commerce, transactions, etc.

 - ⌘ Science: Remote sensing, bioinformatics, etc.

 - ⌘ Society and everyone: news, YouTube, etc.

- ⌘ **Problem:** We are drowning in data, but starving for knowledge!

- ⌘ **Solution:** Use Data Mining tools for Automated Analysis of massive data sets

What Data Mining is Used For? Some Insights

Financial Data Analysis

- ⌘ Banks and Institutions offer a wide variety of banking services
 - ⌘ Checking and saving accounts for business or individual customers
 - ⌘ Credit business, mortgage, and automobile loans
 - ⌘ Investment services (mutual funds)
 - ⌘ Insurance services and stock investment services
- ⌘ Financial data is relatively complete, reliable, and of high quality
- ⌘ What to do with this data?



What Data Mining is Used For? Some Insights

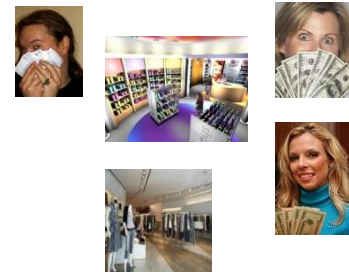
Financial Data Analysis

- ②② Loan Payment Prediction and costumer credit policy analysis
 - ②② Attribute selection and attribute relevance ranking may help identifying important factors and eliminate irrelevant ones
 - ②② Example of factors related to the risk of loan payment
 - ②② Term of the loan
 - ②② Debt ratio
 - ②② Payment to income ratio
 - ②② Customer level income
 - ②② Education level
 - ②② Residence region
- ②② The bank can adjust its decisions according to the subset of factors selected

What Data Mining is Used For? Some Insights

Retail Industry

- Collect huge amount of data on sales, customer shopping history, goods transportation, consumption and service, etc.
- Many stores have web sites where you can buy online. Some of them exist only online (e.g., Amazon)
- Data mining helpsto
 - Identify costumer buying behaviors
 - Discover customers shopping patterns and trends
 - Improve the quality of costumer service
 - Achieve better costumer satisfaction
 - Design more effective good transportation
 - Reduce the cost of business



What Data Mining is Used For? Some Insights

⌘ Many different ways of communicating

⌘ Fax, cellular phone, Internet messenger, images, e-mail, computer and Web data transmission, etc.

⌘ Great demand of data mining to help

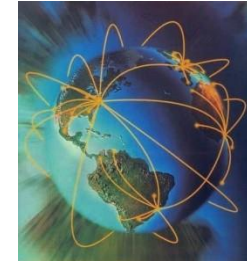
⌘ Understanding the business involved

⌘ Identifying telecommunication patterns

⌘ Catching fraudulent activities

⌘ Making better use of resources

⌘ Improve the quality of service



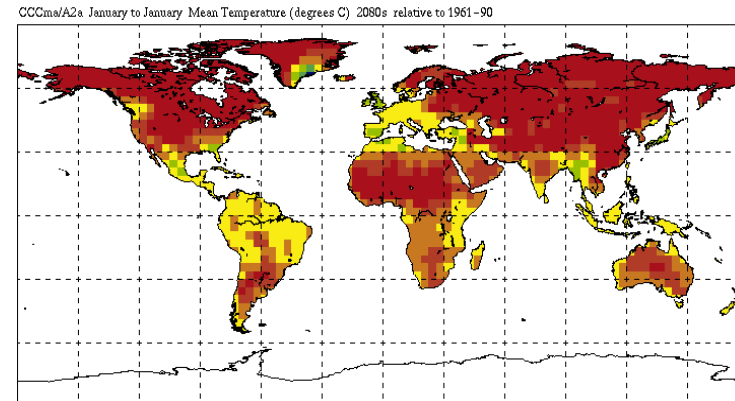
Example of a Data Mining Problem

- ⊗ You want to do advertisement of sport activities for a set of new users on Facebook
- ⊗ These people do not have explicit information about what they like and no history of past activities
- ⊗ What you know is:
 - ⊗ Their age, gender, and location
 - ⊗ Their friends (not necessarily new users)
 - ⊗ Messages they exchange with friends

Great Opportunities to Solve Society's Major Problems



Improving health care and reducing costs



Predicting the impact of climate change

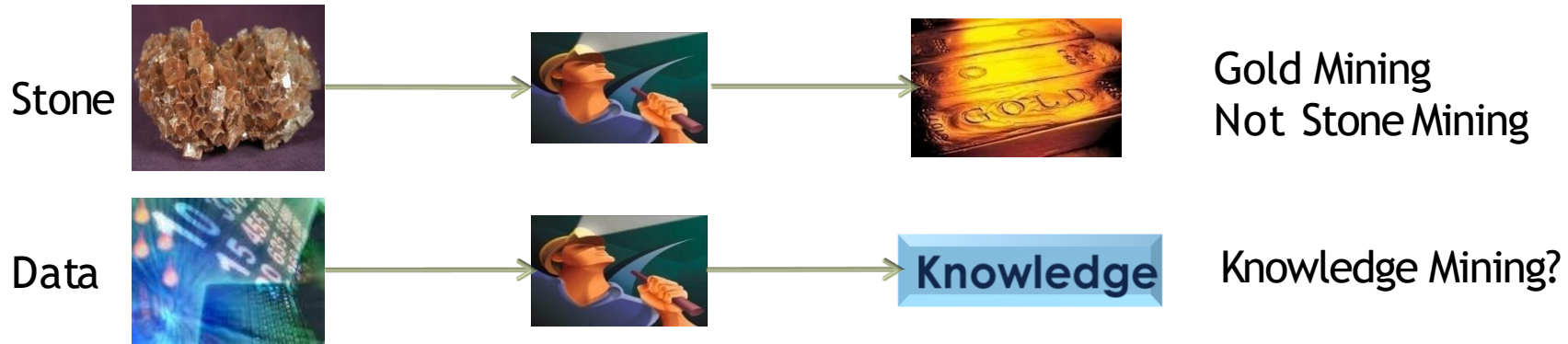


Finding alternative/ green energy sources



Reducing hunger and poverty by increasing agriculture production

Data Mining: what does it?



🔗 Data mining (knowledge discovery from data)

🔗 Extraction of interesting (non-trivial, implicit, previously unknown and potentially useful) patterns or knowledge from huge amount of data

🔗 Alternative names

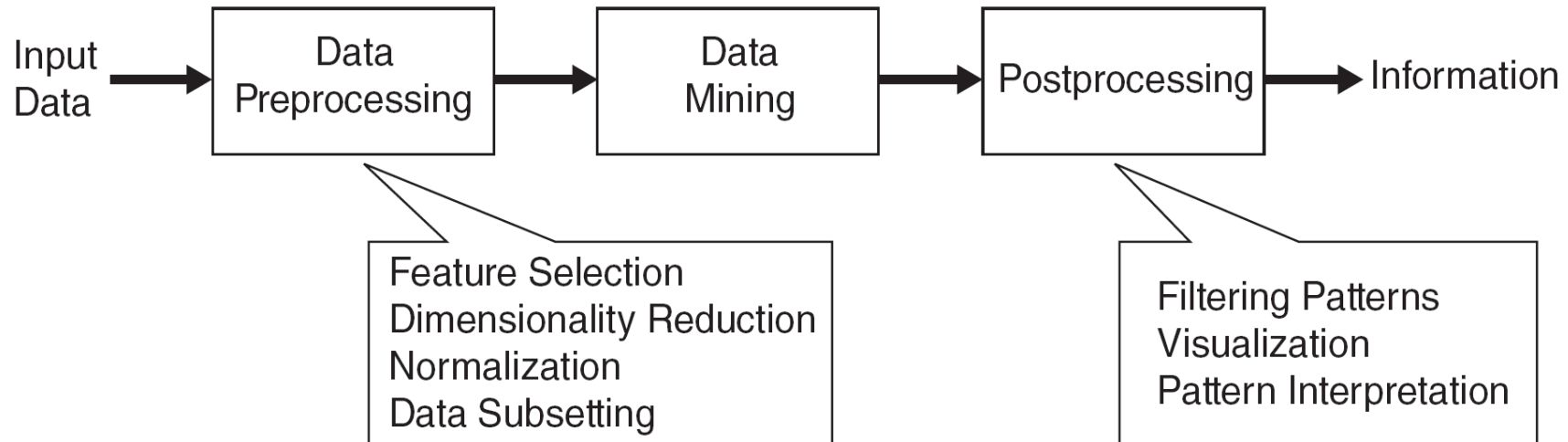
🔗 Knowledge discovery (mining) in databases (KDD), knowledge extraction, data/pattern analysis, data archeology, data dredging, information harvesting, business intelligence, etc.

What is Data Mining?

Many Definitions

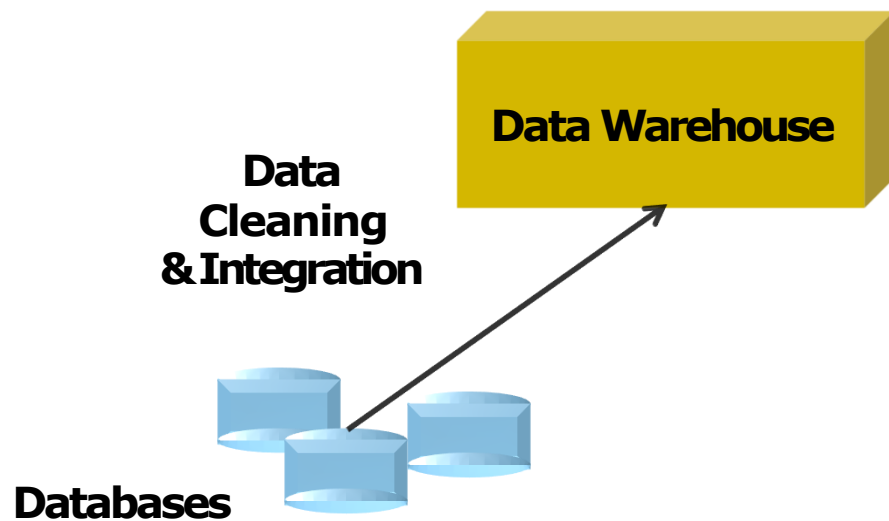
Non-trivial extraction of implicit, previously unknown and potentially useful information from data

Exploration & analysis, by automatic or semi-automatic means, of large quantities of data in order to discover meaningful patterns



Knowledge Discovery (KDD) Process

- 🔗 Data Mining as a step in the knowledge discovery process

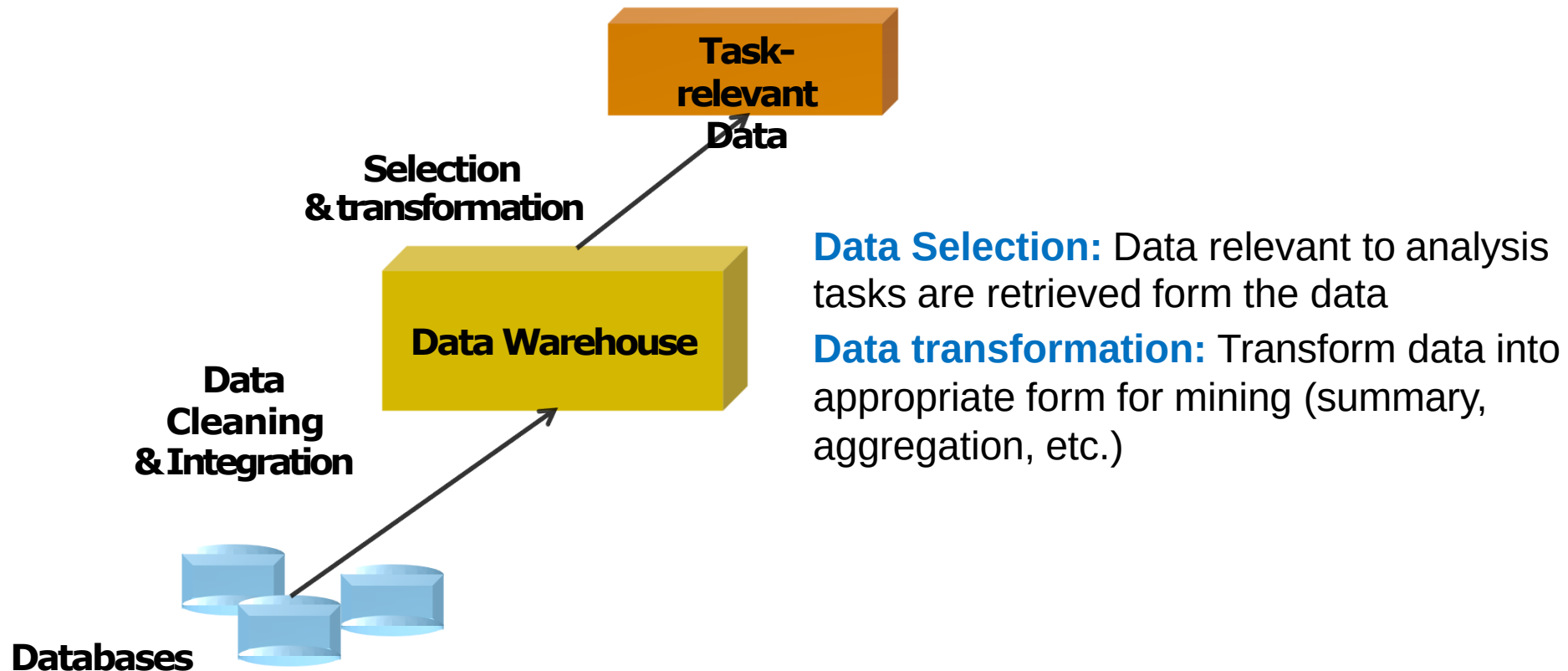


Data Cleaning: Remove noise and inconsistent data

Data Integration: Combine multiple data sources

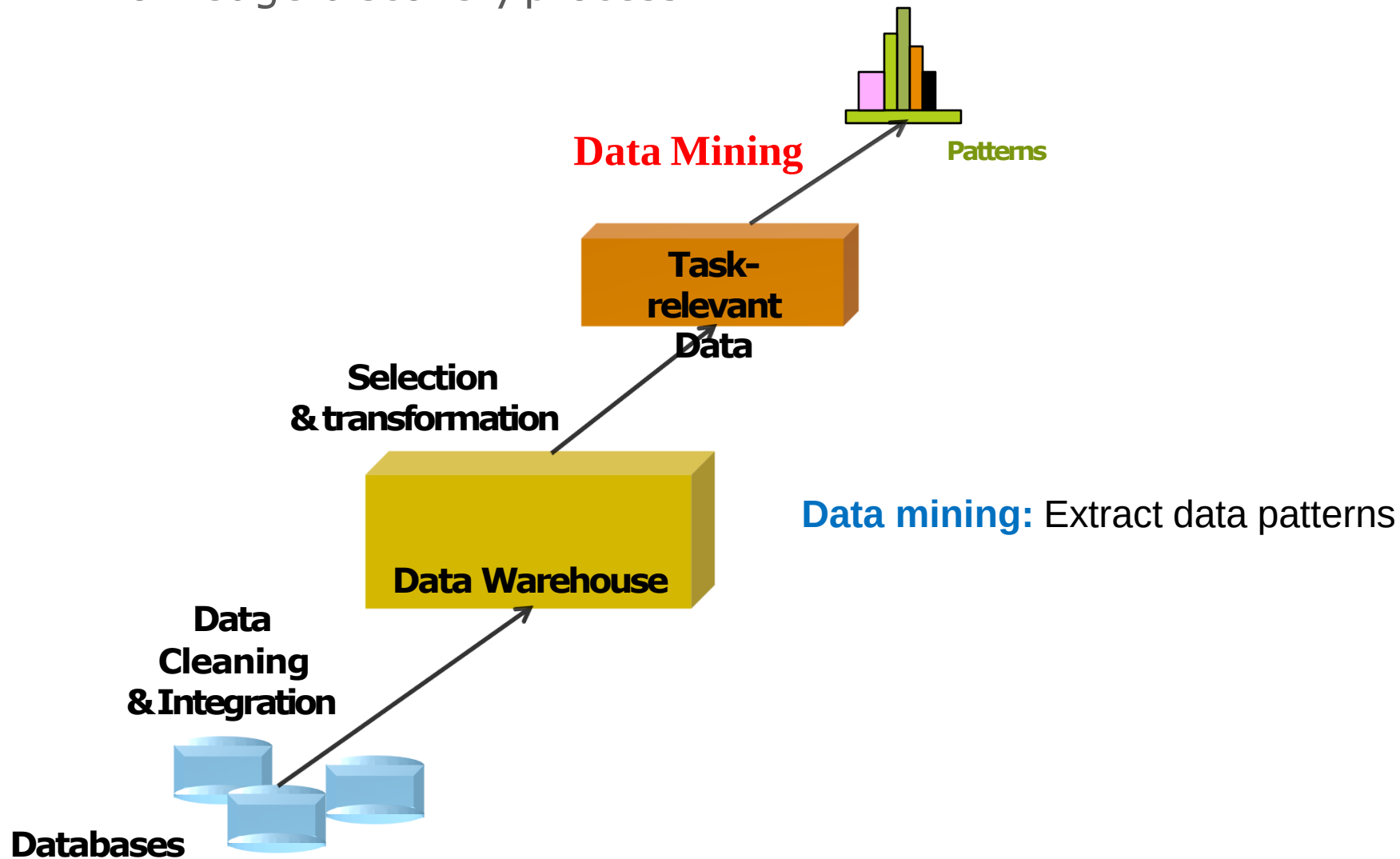
Knowledge Discovery (KDD) Process

- 🔗 Data Mining as a step in the knowledge discovery process



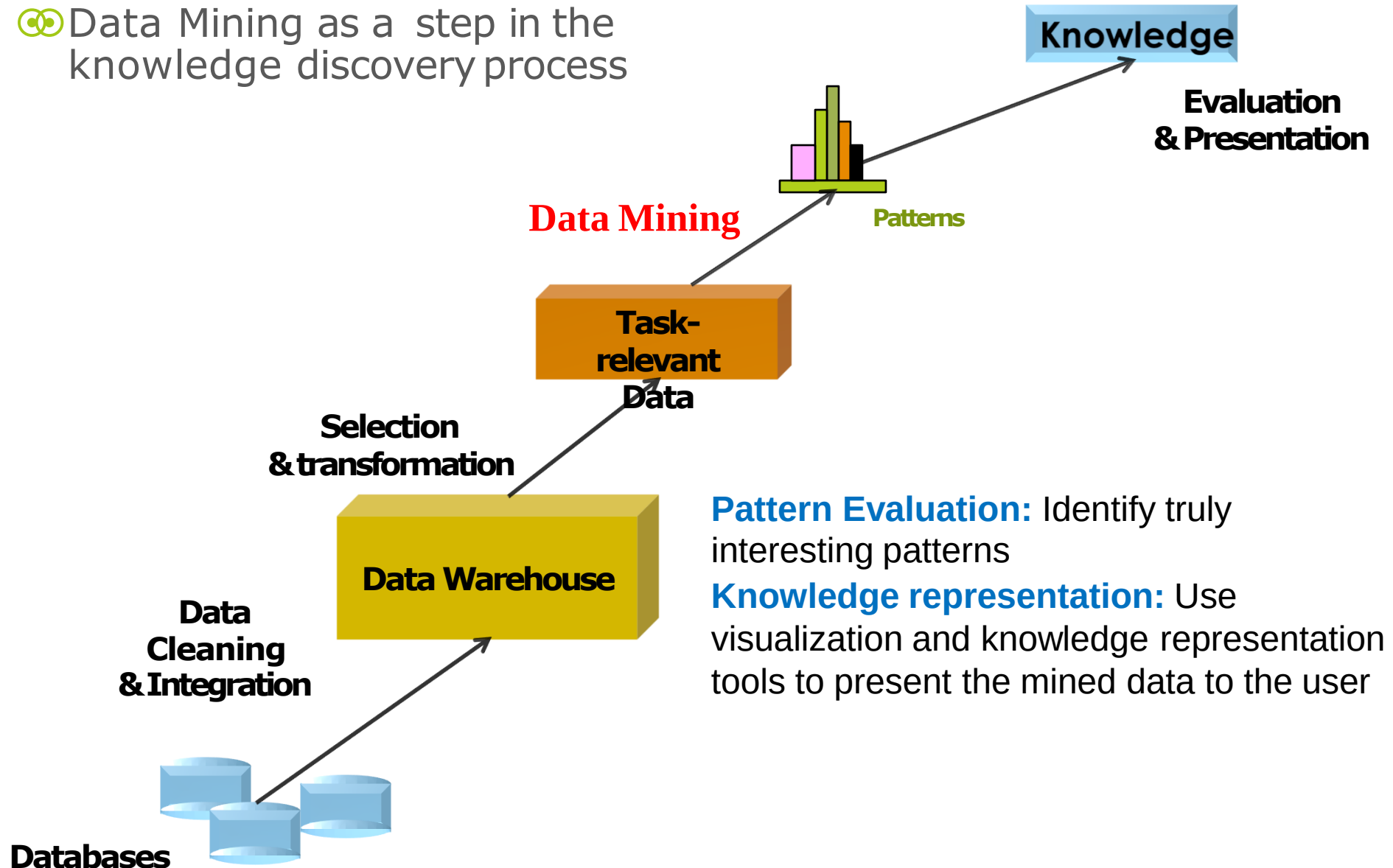
Knowledge Discovery (KDD) Process

🔗 Data Mining as a step in the knowledge discovery process



Knowledge Discovery (KDD) Process

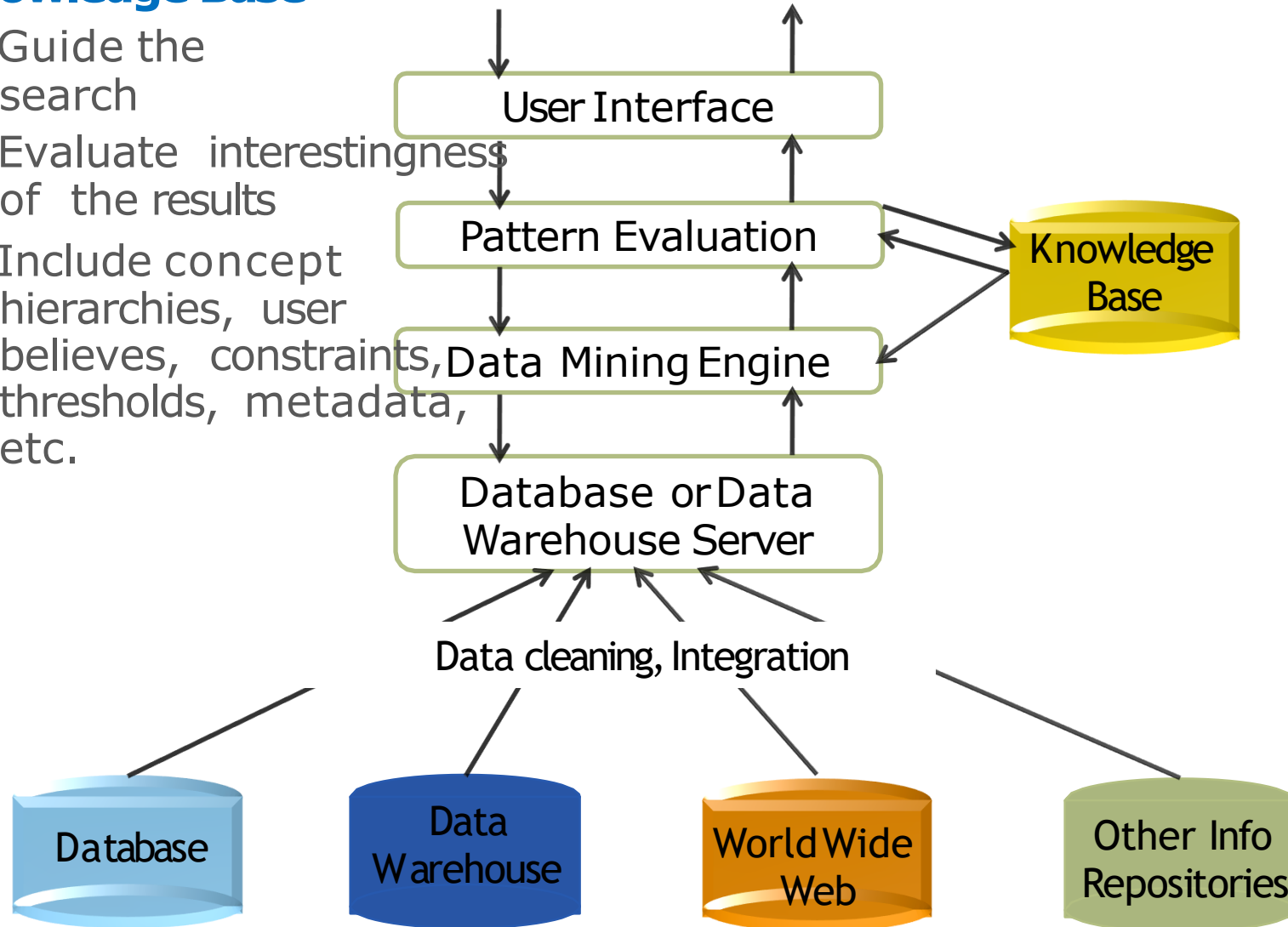
🔗 Data Mining as a step in the knowledge discovery process



Typical Architecture of a DM System

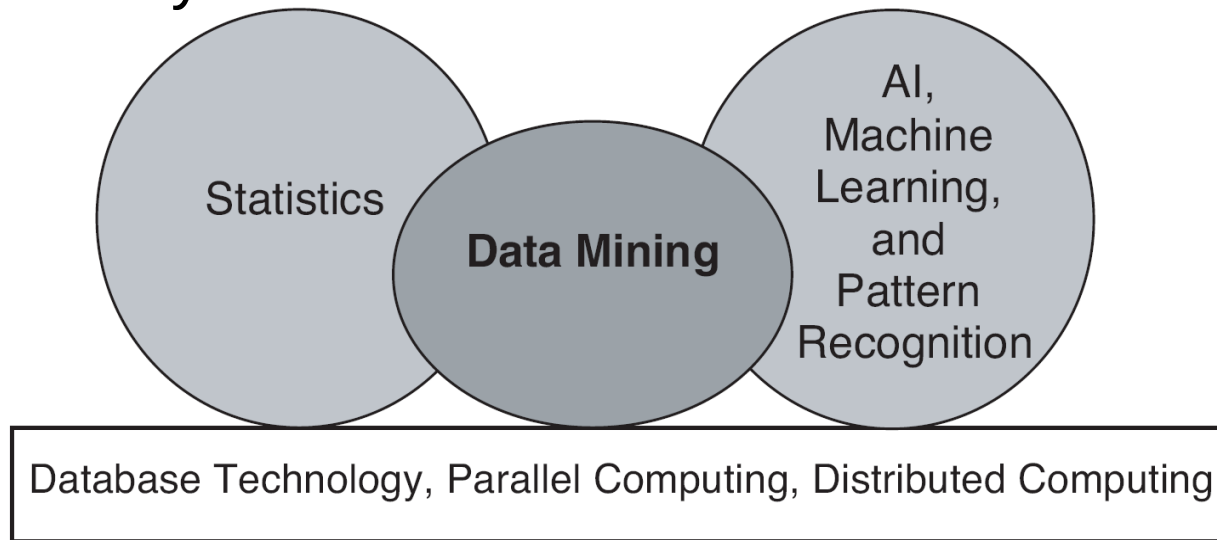
Knowledge Base

- Guide the search
- Evaluate interestingness of the results
- Include concept hierarchies, user believes, constraints, thresholds, metadata, etc.



Origins of Data Mining

- ❑ Draws ideas from machine learning/AI, pattern recognition, statistics, and database systems
- ❑ Traditional techniques may be unsuitable due to data that is
 - Large-scale
 - High dimensional
 - Heterogeneous
 - Complex
 - Distributed
- ❑ A key component of the emerging field of data science and data-driven discovery



Data Mining vs Machine Learning vs Artificial Intelligence



What is Machine Learning?

- ❑ Machine Learning is a technique which develops complex algorithms for processing large data and delivers results to its users. It uses complex programs which can learn through experience and make predictions.
- ❑ The algorithms are improved by itself through regular input of training data. The goal of machine learning is to understand data and build models from data that can be understood and used by humans.

3 Categories of Machine Learning

Supervised Learning:

In supervised learning, both the input and desired output are provided and the machine must learn how to map the former to the latter. To accomplish this, the machine is trained on a statistically representative set of example inputs and corresponding outputs.

3 Categories of Machine Learning

Unsupervised Learning:

In unsupervised learning, the machine is not provided labeled examples or previous patterns on which to base the analysis of the data inputs. The machine must uncover patterns and draw inferences by itself, without having the correct answers. It will classify or cluster data by discovering the similarity of features on its own.

3 Categories of Machine Learning

Reinforcement Learning:

Reinforcement learning differs from both supervised and unsupervised learning by continuously improving its model based on feedback from experiences. It learns through trial and error – from the consequences of its action and by new choices. As an action is taken, the success of the outcome is graded and receives either a positive or negative score. The algorithm seeks to receive positive scores and the model is trained on continuous feedback.

Artificial Intelligence and Data Mining

Artificial Intelligence is the study to create intelligent machines which can work like humans. It does not depend on learning or feedback, rather it has directly programmed control systems. The AI systems come up with the solutions to the problems on their own by calculations.

The data mining technique in mined data is used by the AI systems for creating solutions. Data mining serves as a foundation for artificial intelligence. Data mining is a part of programming codes with information and data necessary for AI systems.

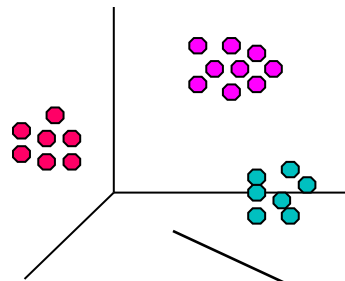
Data Mining Tasks

- Prediction Methods
 - Use some variables to predict unknown or future values of other variables.

- Description Methods
 - Find human-interpretable patterns that describe the data.

From [Fayyad, et.al.] Advances in Knowledge Discovery and Data Mining, 1996

Data Mining Tasks ...



Clustering

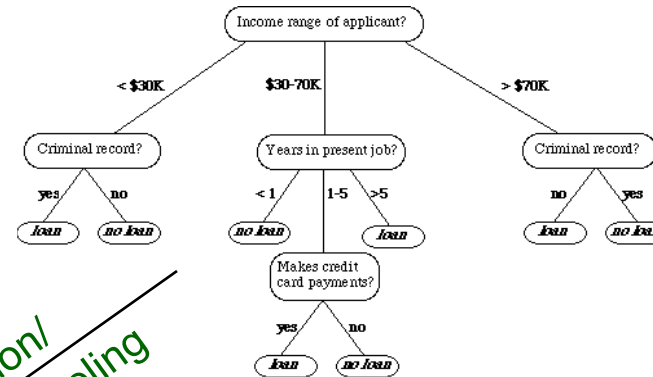
Data

Tid	Refund	Marital Status	Taxable Income	Cheat
1	Yes	Single	125K	No
2	No	Married	100K	No
3	No	Single	70K	No
4	Yes	Married	120K	No
5	No	Divorced	95K	Yes
6	No	Married	60K	No
7	Yes	Divorced	220K	No
8	No	Single	85K	Yes
9	No	Married	75K	No
10	No	Single	90K	Yes
11	No	Married	60K	No
12	Yes	Divorced	220K	No
13	No	Single	85K	Yes
14	No	Married	75K	No
15	No	Single	90K	Yes

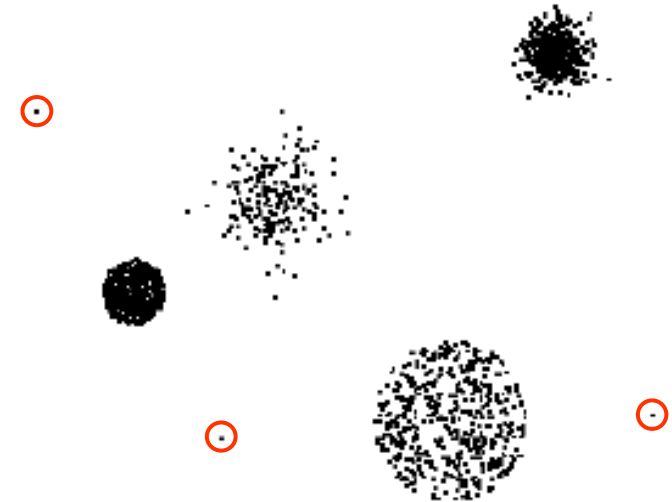
Association
Classification



Regression/
Predictive Modeling



Anomaly
Detection

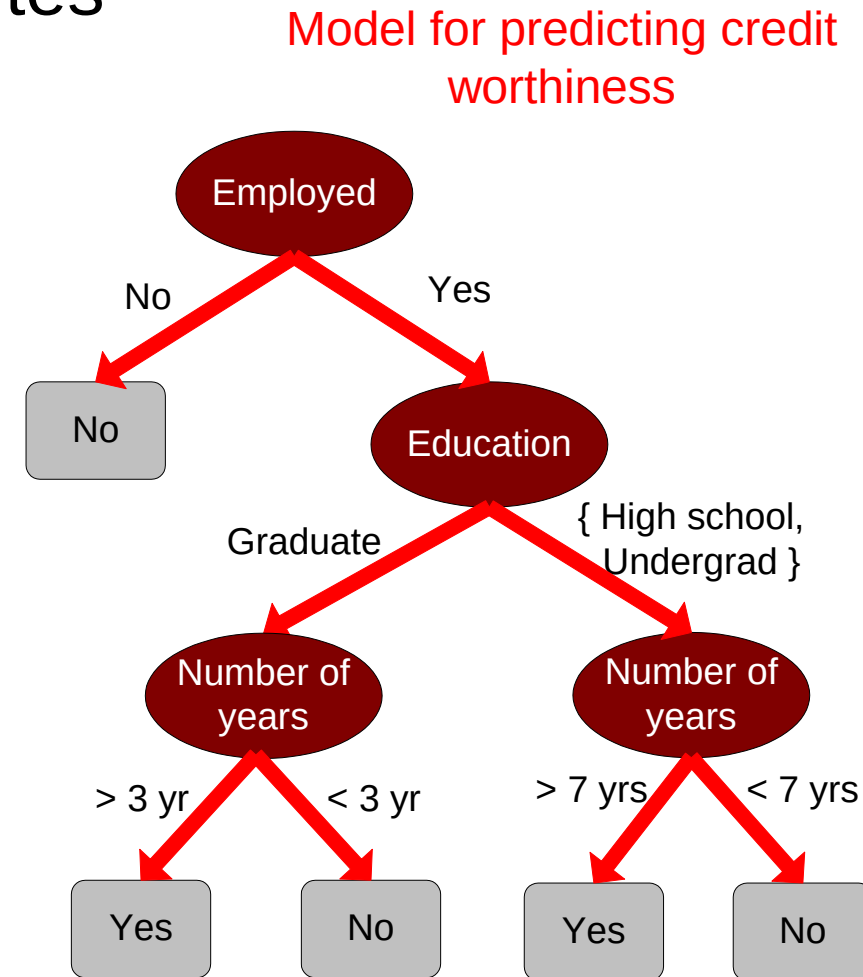


Predictive Modeling: Classification

- Find a model for class attribute as a function of the values of other attributes

Class

<i>Tid</i>	Employed	Level of Education	# years at present address	Credit Worthy
1	Yes	Graduate	5	Yes
2	Yes	High School	2	No
3	No	Undergrad	1	No
4	Yes	High School	10	Yes
...	...	...	...	...

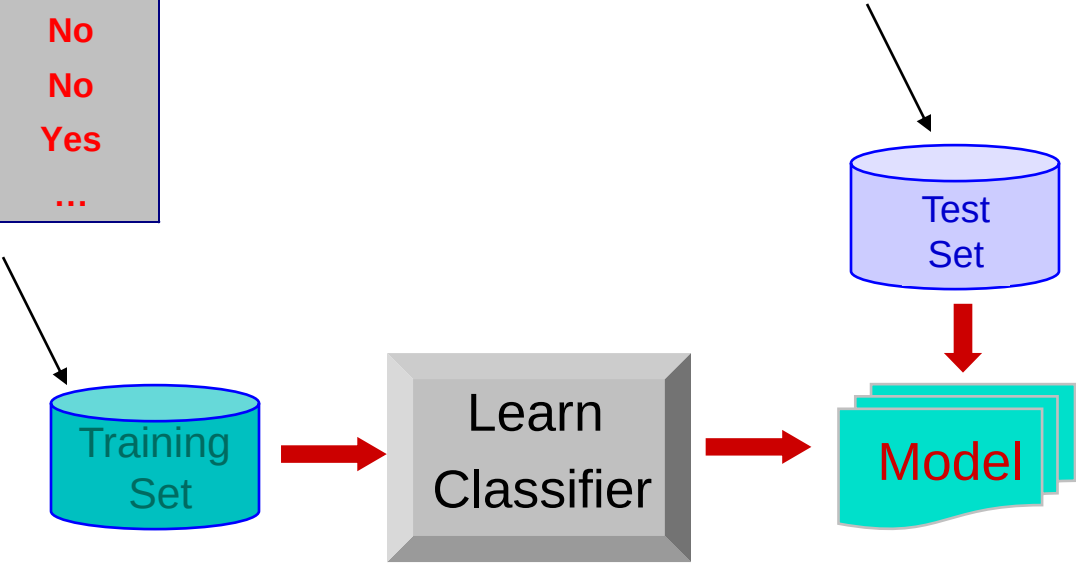


Classification Example

categorical categorical quantitative class

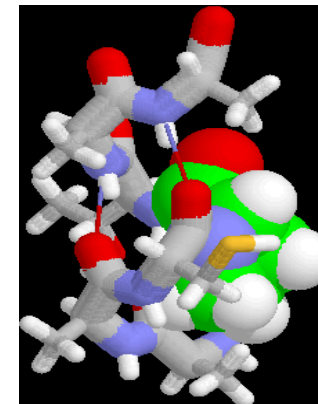
<i>Tid</i>	Employed	Level of Education	# years at present address	Credit Worthy
1	Yes	Graduate	5	Yes
2	Yes	High School	2	No
3	No	Undergrad	1	No
4	Yes	High School	10	Yes
...	...	...	...	...

<i>Tid</i>	Employed	Level of Education	# years at present address	Credit Worthy
1	Yes	Undergrad	7	?
2	No	Graduate	3	?
3	Yes	High School	2	?
...	...	...	...	...



Examples of Classification Task

- ❑ Classifying credit card transactions as legitimate or fraudulent
- ❑ Classifying land covers (water bodies, urban areas, forests, etc.) using satellite data
- ❑ Categorizing news stories as finance, weather, entertainment, sports, etc
- ❑ Identifying intruders in the cyberspace
- ❑ Predicting tumor cells as benign or malignant
- ❑ Classifying secondary structures of protein as alpha-helix, beta-sheet, or random coil

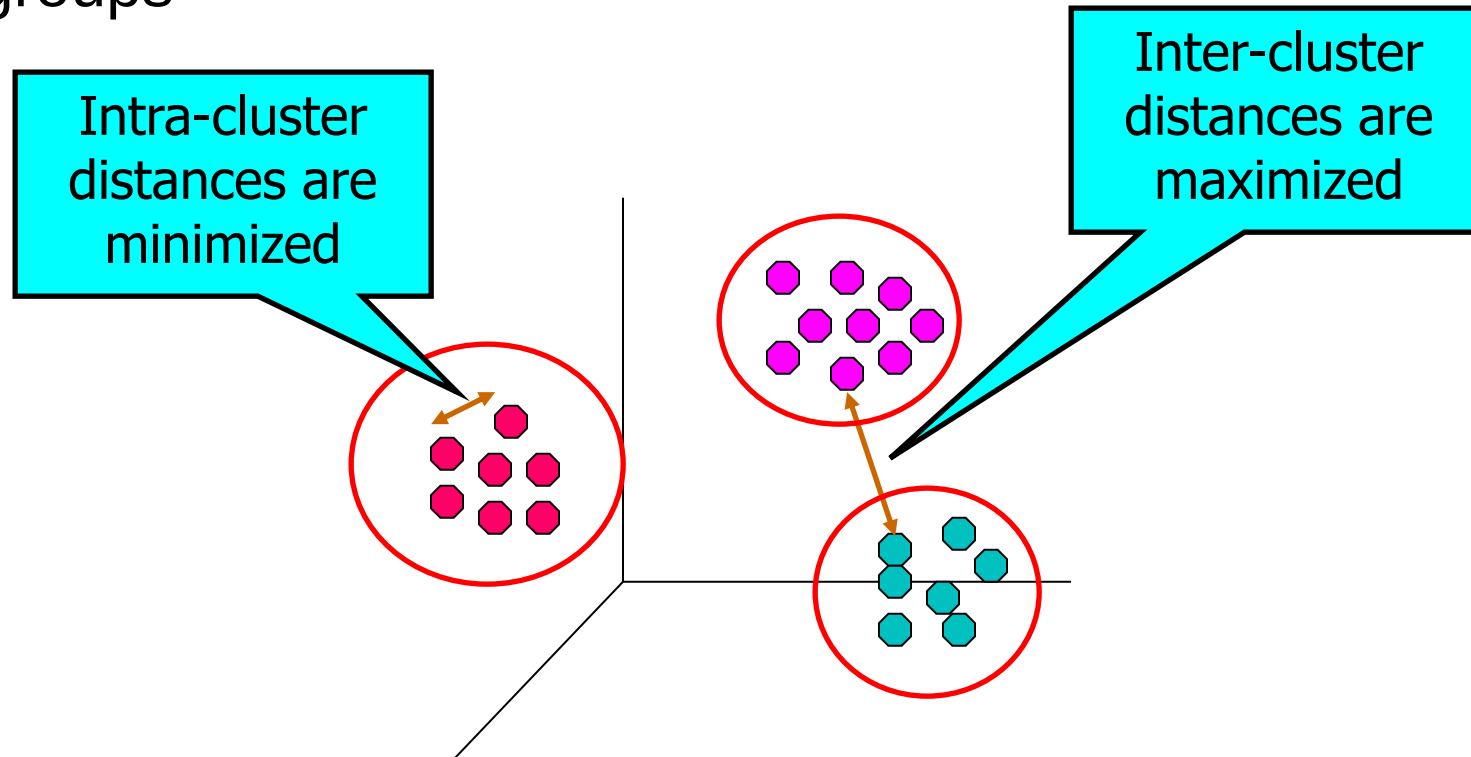


Regression

- Predict a value of a given continuous valued variable based on the values of other variables, assuming a linear or nonlinear model of dependency.
- Extensively studied in statistics, neural network fields.
- Examples:
 - Predicting sales amounts of new product based on advertising expenditure.
 - Predicting wind velocities as a function of temperature, humidity, air pressure, etc.
 - Time series prediction of stock market indices.

Clustering

- Finding groups of objects such that the objects in a group will be similar (or related) to one another and different from (or unrelated to) the objects in other groups



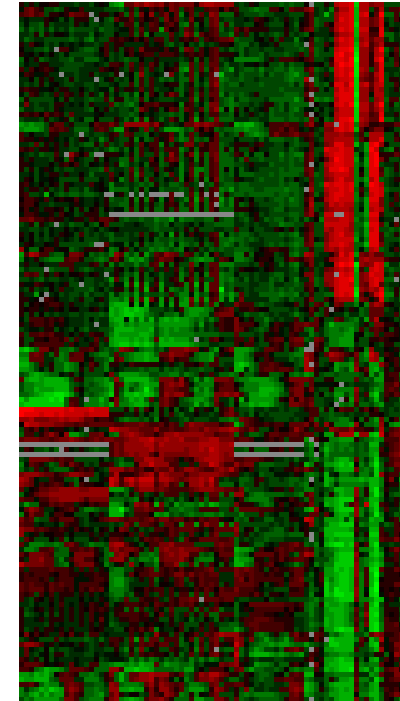
Applications of Cluster Analysis

□ Understanding

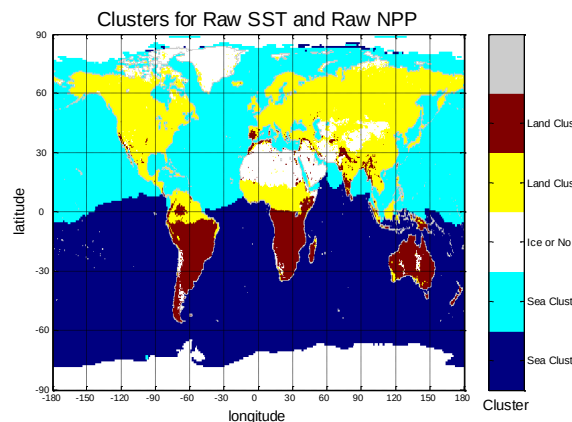
- Custom profiling for targeted marketing
- Group related documents for browsing
- Group genes and proteins that have similar functionality
- Group stocks with similar price fluctuations

□ Summarization

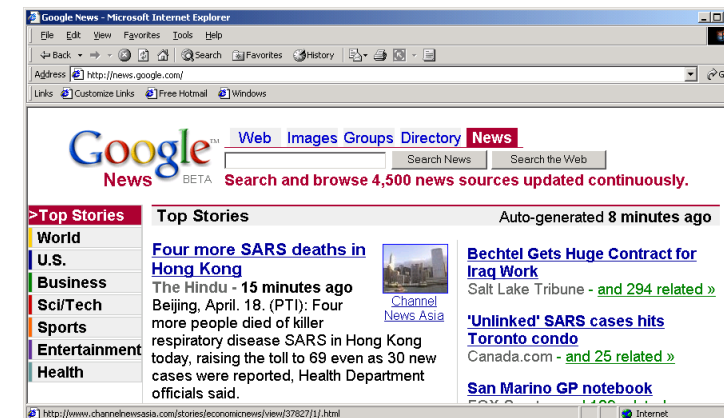
- Reduce the size of large data sets



Courtesy: Michael Eisen



Use of K-means to partition Sea Surface Temperature (SST) and Net Primary Production (NPP) into clusters that reflect the Northern and Southern Hemispheres.



Association Rule Discovery: Definition

- Given a set of records each of which contain some number of items from a given collection
 - Produce dependency rules which will predict occurrence of an item based on occurrences of other items.

<i>TID</i>	<i>Items</i>
1	Bread, Coke, Milk
2	Beer, Bread
3	Beer, Coke, Diaper, Milk
4	Beer, Bread, Diaper, Milk
5	Coke, Diaper, Milk

Rules Discovered:

$\{\text{Milk}\} \rightarrow \{\text{Coke}\}$

$\{\text{Diaper, Milk}\} \rightarrow \{\text{Beer}\}$

Association Analysis: Applications

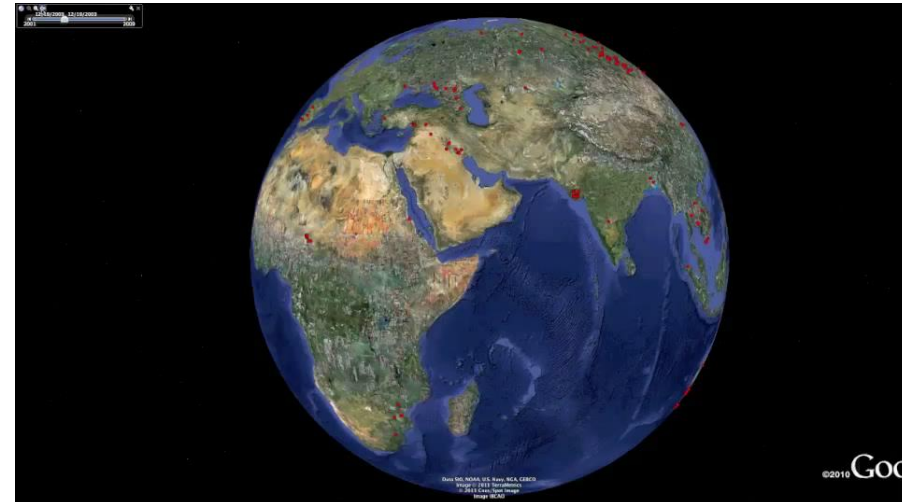
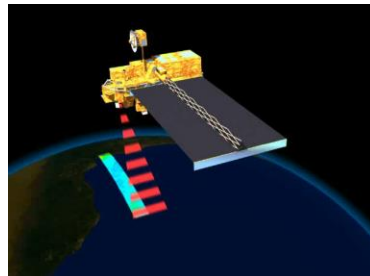
- Market-basket analysis
 - Rules are used for sales promotion, shelf management, and inventory management

- Telecommunication alarm diagnosis
 - Rules are used to find combination of alarms that occur together frequently in the same time period

- Medical Informatics
 - Rules are used to find combination of patient symptoms and test results associated with certain diseases

Deviation/Anomaly/Change Detection

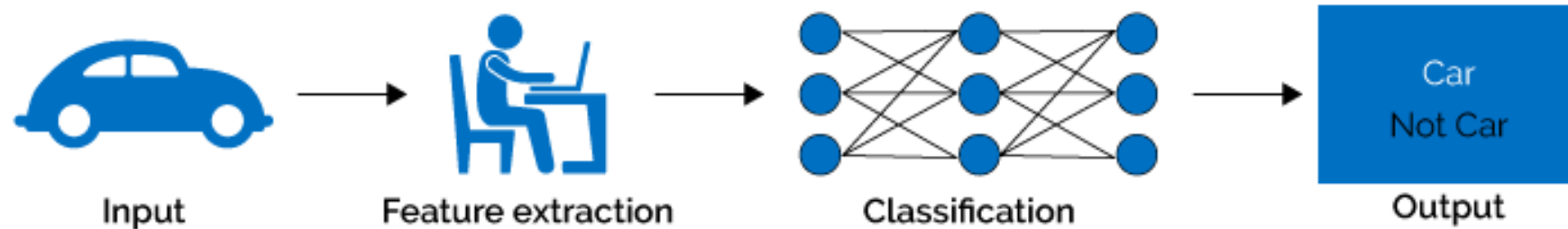
- Detect significant deviations from normal behavior
- Applications:
 - Credit Card Fraud Detection
 - Network Intrusion Detection
 - Identify anomalous behavior from sensor networks for monitoring and surveillance.
 - Detecting changes in the global forest cover.



Deep Learning

- ❑ Deep Learning is a subset of machine learning. It works in the same way on the machine just like how the human brain processes information. Like a brain can identify the patterns by comparing it with previously memorized patterns, deep learning also uses this concept.
- ❑ Deep learning can automatically find out the attributes from raw data while machine learning selects these features manually which further needs processing. It also employs artificial neural networks with many hidden layers, big data, and high computer resources.

Machine Learning



Deep Learning

